

Uncertainty: completing the EnKF evaluation, and a third method

Both tasks done — 7-day EnKF evaluation at $k=1$ and $k=4$, and deep ensembles implemented and compared

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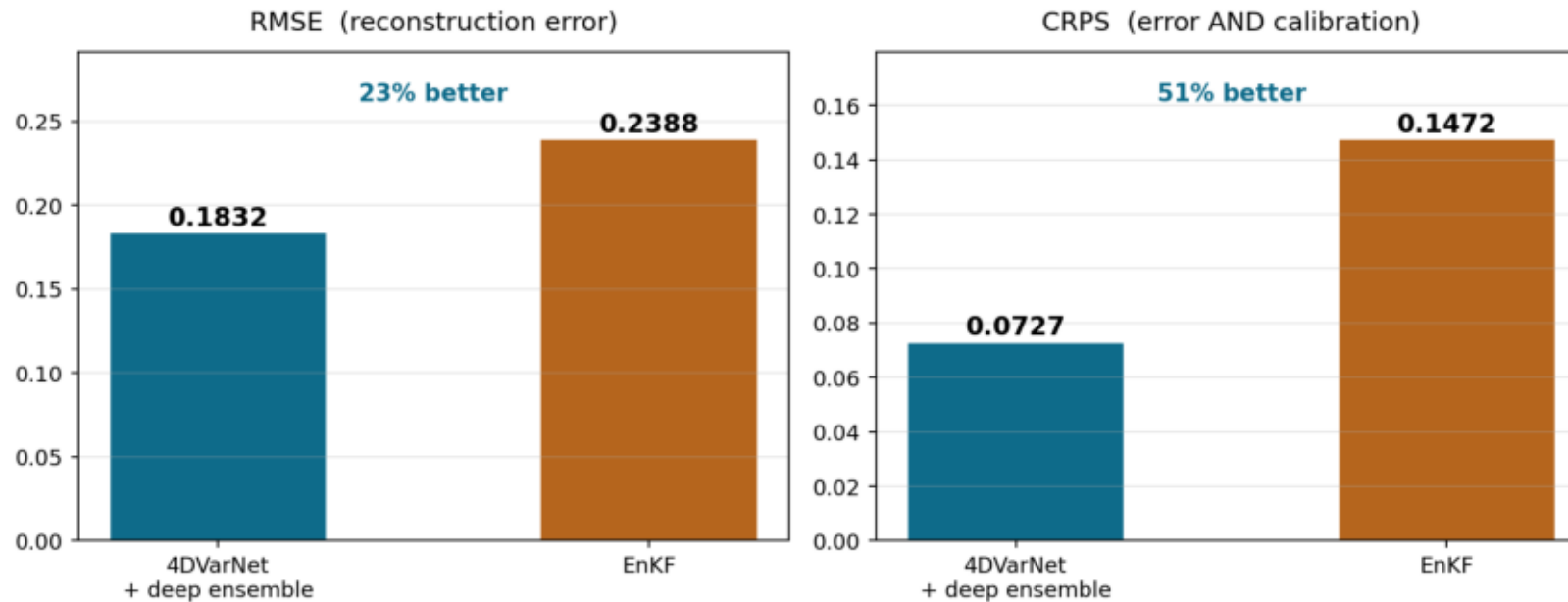
Task 1 — the optimised EnKF, 7 days, both observation densities

- Accuracy and uncertainty both complete, on all 7 held-out test days (479,594,304 points per setting).
 - k=1 (every frame observed) RMSE 0.2388 CRPS 0.1472 spread/skill 0.011 90% coverage 2.24%
 - k=4 (every 4th frame) RMSE 0.2428 CRPS 0.1475 spread/skill 0.010 90% coverage 1.99%
- Four times the observations changes almost nothing: RMSE moves 1.7%, CRPS 0.2%. Together with the Kalman correction accounting for 0.22% of the forecast error, this says the observations are barely entering the state.

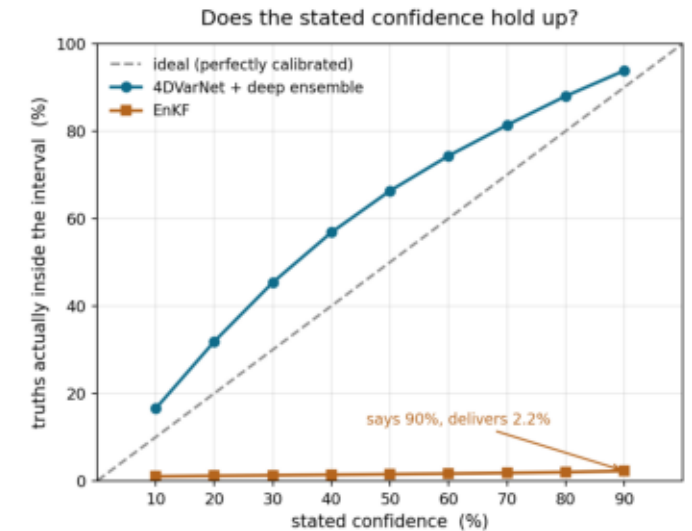
Task 2 — the third method: deep ensembles

- Lakshminarayanan, Pritzel & Blundell, NeurIPS 2017 (arXiv:1612.01474). Chosen because it is the standard against which uncertainty methods are measured, and because it needs no change to the reconstruction itself.
- The recipe, and what we implemented
 - each network predicts a mean AND a variance, and is trained on the Gaussian NLL instead of MSE (their Eq.1) — a proper scoring rule, so it rewards a calibrated sigma rather than merely a small error
 - 5 members, differing only in initialisation (5 trained, 150 epochs each, ~79 GPU-hours); combined by their Sec. 2.4 moment matching
 - adversarial training (their Sec. 2.3) skipped: optional, and their Table 2 states it does not help on regression
- The variational cost $J(x)$ is untouched — only the training loss changed, which is a substitution 4DVarNet explicitly allows. The solver still descends the gradient of the cost, computed from observations alone.

Result — better on accuracy and on uncertainty



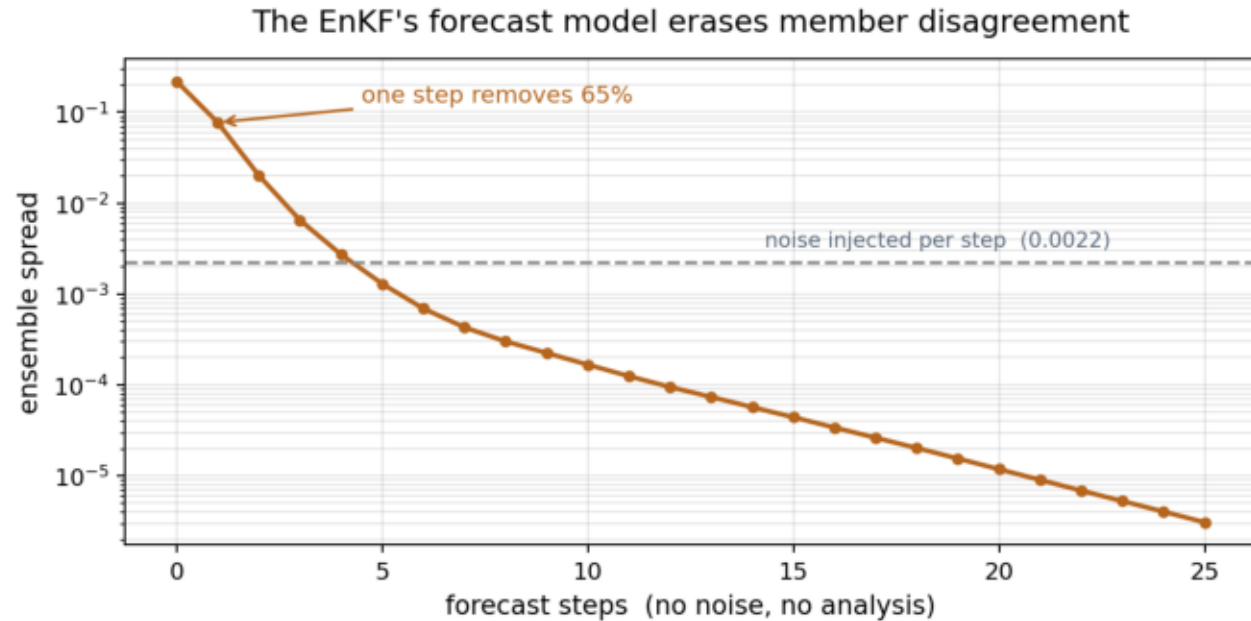
7 held-out test days, identical observations and evaluation frames for both methods. CRPS is a proper scoring rule in the units of the state: it penalises a wrong mean and a wrong spread at once, so a method cannot score well by being merely accurate or merely uncertain. NLL is reported in the tables but not drawn -- the EnKF's is $-1e18$ because its sigma approaches zero.



Protocol from Lakshminarayanan et al. App. A.2. A point on the diagonal means the method's stated confidence is exactly right. Above it is conservative (intervals wider than needed); below it is overconfident. The EnKF's curve is nearly flat: widening its interval 9x moves coverage only from 1.00% to 2.24%, so its spread barely tracks its error and no rescaling would fix it.

- RMSE 0.1832 against 0.2388 (23% better); CRPS 0.0727 against 0.1472 (51% better).
- Calibration: spread/skill 0.87 against 0.011 (1.0 is perfect), and 0.99 in the blind cells, which is where it has to work. The observed cells are over-confident at 0.71 — we are not uniformly on the safe side.
- Cost of the change: full-state RMSE 0.1599 -> 0.1616, +1.1% at matched epochs. The paper reports a median penalty of 5.6% for the same trade.

Why the EnKF's uncertainty fails — and why tuning would not fix it



An ensemble of 100 members is given a spread of 0.218 -- the full configured process noise, 100x what the filter actually injects -- and then advanced through the forecast model alone: no process noise, no Kalman analysis, no inflation. The spread decays to 3.1e-06 in 25 steps. The model is contractive, so the collapse is not a matter of retuning: raising the injected noise or the inflation would buy spread only by degrading the forecast.

- Its uncertainty IS the ensemble spread, and the forecast model removes member disagreement faster than any noise term can restore it.
- So the collapse is structural, not a bad setting: a deterministic surrogate trained to minimise prediction error is trained to be contractive, which is exactly what an ensemble needs it not to be.
- This is why we did not retune the baseline. Raising the inflation or the process noise buys spread only by corrupting the forecast.

Where this leaves us

- Settled

- the EnKF evaluation is complete at both observation densities, and the comparison is favourable on every metric we measured
- the uncertainty is calibrated and it is informative — it is larger in cells no robot saw (0.196) than in observed ones (0.118), and it beats a constant-sigma null model on CRPS (0.0727 against 0.0861)

- Open

- the observed cells are over-confident at 0.71 spread/skill while the blind cells sit at 0.99. The heads were fitted WITH a σ^2 floor and are being read without one; a floorless retrain (runs/varnet_nf5_s*) is running now and should say whether that split is the cause
- we have no 4DVarNet uncertainty at $k=4$; given that the EnKF barely moves between $k=1$ and $k=4$, the value of adding it looks low against 79 more GPU-hours
- is the three-way benchmark meant to compare three reconstruction methods, or three uncertainty treatments? That decides the next search